

Unit 2 Progress Check: FRQ

Name _____

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about N_2 and N_2H_4 .

- (a) In the box below, draw the complete Lewis electron-dot diagram of N_2 .



Please respond on separate paper, following directions from your teacher.

- (b) Based on the Lewis electron-dot diagram that you drew, is the N_2 molecule polar? Explain.

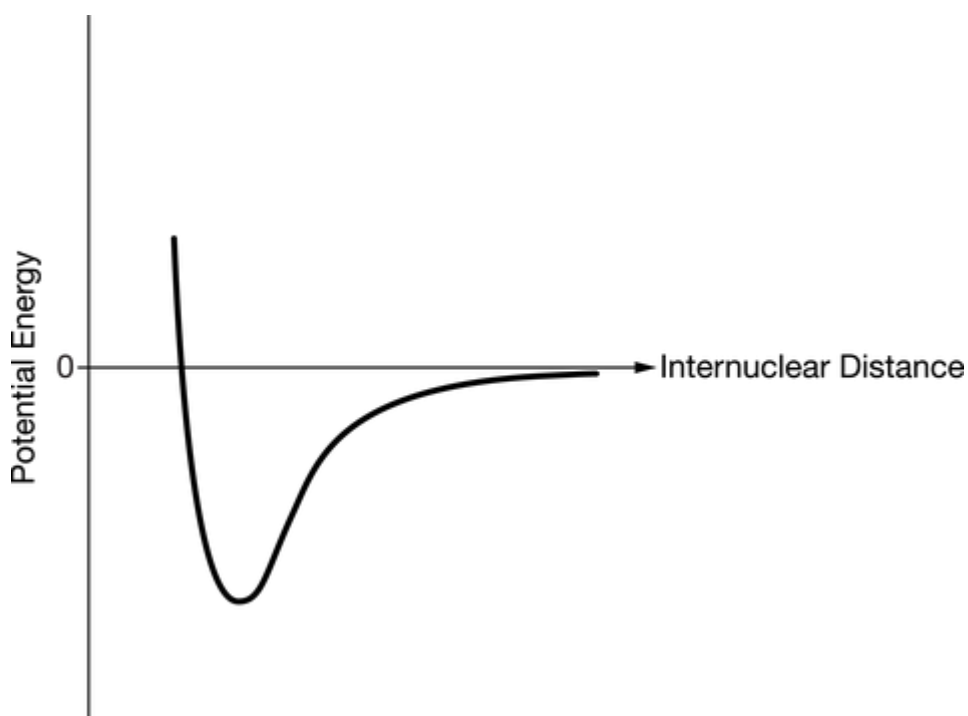


Please respond on separate paper, following directions from your teacher.

The following graph shows the potential energy of two nitrogen atoms versus the distance between their nuclei.



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(c) On the graph, indicate the distance that corresponds to the bond length of the N_2 molecule by placing an **X** on the horizontal axis.



Please respond on separate paper, following directions from your teacher.

(d) On the graph, which shows the potential energy curve of two **N** atoms, carefully sketch a curve that corresponds to the potential energy of two **O** atoms versus the distance between their nuclei.



Please respond on separate paper, following directions from your teacher.

N_2 can react with H_2 to form the compound N_2H_4 .

(e) A sample of N_2H_4 has a mass of **25 g**. Identify the numerical quantity that is needed to convert the number of grams of N_2H_4 to the number of moles of N_2H_4 . (You do not need to do the actual calculation.)

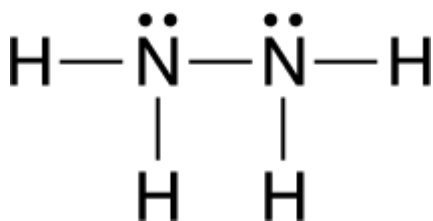


Please respond on separate paper, following directions from your teacher.

(f) The Lewis electron-dot diagram of N_2H_4 is shown below.



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- (i) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4 , compare the length of the nitrogen-to-nitrogen bond in N_2 with the length of the nitrogen-to-nitrogen bond in N_2H_4 .



Please respond on separate paper, following directions from your teacher.

- (ii) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4 , compare the strength of the nitrogen-to-nitrogen bond in N_2 with the strength of the nitrogen-to-nitrogen bond in N_2H_4 .



Please respond on separate paper, following directions from your teacher.

- (iii) Identify the hybridization of the N atoms in N_2H_4 .



Please respond on separate paper, following directions from your teacher.